



Name: _____

Class: _____

Date: _____

Mr. Rawson • IB DP Physics

Simple Harmonic Motion 4 – Analysing Your Oscillation Data

Answer Key

You already have the numbers. This lesson turns your spring–mass results into a spring constant, checks your prediction against your own measurement, and uses your own oscillator to work through how displacement, velocity and acceleration relate. Keep your practical packet and spreadsheet open — every question below uses *your* data, not a supplied one.

Linearizing: from your graph to a spring constant

Key concept

$T = 2\pi\sqrt{m/k}$ is a square-root relationship — a graph of T against m curves, and a curve's gradient changes from point to point, which makes it useless for finding k . Squaring both sides straightens it:

$$T^2 = \frac{4\pi^2}{k} m$$

T^2 against m is now a straight line through the origin (plus a small manufacturing offset, more below), with gradient $4\pi^2/k$. This is **linearizing**: choosing the version of a relationship whose graph is straight, because a straight line is the one shape whose gradient is a single, measurable number.

- 1) **State** why plotting T^2 against m lets you find k , when plotting T against m does not.

[2]

Mark scheme

T against m is a curve, and a curve has no single gradient to measure (1). Squaring T turns the relationship into a straight line, $T^2 = (4\pi^2/k)m$, whose gradient is one fixed number equal to $4\pi^2/k$ (1).

Worked example — finding k from a gradient

A group's T^2 vs m graph has gradient $0.51 \text{ s}^2/\text{kg}$.

$$\text{gradient} = \frac{4\pi^2}{k} \Rightarrow k = \frac{4\pi^2}{\text{gradient}} = \frac{4\pi^2}{0.51} = 77.4 \text{ N/m}$$

Check: a stiffer spring gives a smaller gradient. If your k comes out smaller than a spring you can barely stretch by hand should have, re-read your gradient before going further.

2) **Calculate** k from the gradient of your own T^2 vs m graph. Show the rearrangement.

[3]

My gradient = _____ s^2/kg

k = _____ N/m

3) **Explain** why your graph has a small positive T^2 -intercept instead of passing through the origin, and state whether it affects the value of k you just calculated.

[3]

Mark scheme

The spring itself has mass, and that mass also oscillates, so the true restoring system is slightly heavier than the hanging mass alone (1). This shifts the whole line up by a small fixed amount without changing its steepness (2). The gradient — and therefore k — is unaffected; only the intercept carries the effect (1).

4) **Compare** your calculated k against a direct measurement: pick one of your six masses, use your k to predict its period, then compare that prediction with the period you actually measured for that mass.

[4]

Mass chosen, m = _____ kg

Measured T = _____ s

Predicted $T = 2\pi\sqrt{m/k}$ = _____ s

Percentage difference between predicted and measured: _____ %

Mark scheme

Full marks requires the correct substitution of the student's own m and k into $T = 2\pi\sqrt{m/k}$ (2), a correctly evaluated predicted T (1), and a percentage-difference calculation $\left(\frac{|T_{\text{pred}} - T_{\text{meas}}|}{T_{\text{meas}}}\right) \times 100\%$ carried through correctly (1). A close agreement (single-digit %) confirms the gradient method; a large one is worth revisiting the graph, not the arithmetic.

What the amplitude check told you

5) **State** the pattern in your three amplitude-check readings — the same mass, three different pull-down distances.

[2]

Mark scheme

The period comes out the same (within experimental uncertainty) for all three pull-down distances (2). Full marks requires the student to have actually compared their own three period values, not asserted the property from memory.

6) **Connect** your amplitude-check result to the formula $T = 2\pi\sqrt{m/k}$. What feature of the formula does your result confirm?

[2]

Mark scheme

The formula contains no amplitude term at all — T depends only on m and k (1). Your result is the experimental confirmation of that: changing how far you pulled the mass down changed nothing about how long each swing took (1).

Position, velocity and acceleration — for your own oscillator

Key concept

Three quantities describe an oscillator at any instant: displacement x , velocity v , and acceleration $a = -\omega^2 x$. Knowing one at a given moment, together with ω , gets you the others. You already have everything needed to find ω for your own spring.

7) **Calculate** ω for your oscillator, using the period you measured for the mass you chose in question 4.

[2]

$$\omega = \frac{2\pi}{T} = \text{_____} \text{ rad/s}$$

Mark scheme

Correct substitution of the student's own measured T into $\omega = 2\pi/T$ (1), correctly evaluated (1).

(a) You pulled the mass down by roughly 2 cm before releasing it — that was your amplitude, x_0 . Using your own ω , calculate the acceleration at the instant of release, when $x = x_0$.

[3]

Mark scheme

$a = -\omega^2 x_0$, with the student's own ω and $x_0 = 0.020$ m (2). The negative sign is required and should be interpreted: at release the object accelerates back *towards* equilibrium, opposite to the displacement (1).

(b) State the acceleration at the instant the mass passes through the equilibrium position, and explain why.

[2]

Mark scheme

Zero (1), because $a = -\omega^2 x$ and at equilibrium $x = 0$ (1).

8) **Complete** the table below for your own oscillator, at the three points in one full cycle listed.

[6]

Point in cycle	Displacement x	Velocity v	Acceleration a
Released, $x = x_0$			
Passing equilibrium			
Other extreme, $x = -x_0$			

Mark scheme

Released ($x = x_0$): $v = 0$ (momentarily at rest); $a = -\omega^2 x_0$ (maximum magnitude, directed back towards equilibrium). *Equilibrium* ($x = 0$): v is maximum, $v = \omega x_0$; $a = 0$. *Other extreme* ($x = -x_0$): $v = 0$; $a = +\omega^2 x_0$ (maximum magnitude, now directed the other way). One mark per correctly reasoned row; the sign pattern — v and a never both maximum together — is the mark examiners look for.

Higher level only

Speed at a chosen displacement.

$$v = \pm\omega\sqrt{x_0^2 - x^2}$$

9) **Calculate** the speed of your own oscillator at the instant its displacement is half its amplitude, $x = x_0/2$. Use your own ω and $x_0 = 0.020$ m.

[4]

Mark scheme

$v = \pm\omega\sqrt{x_0^2 - (x_0/2)^2} = \pm\omega\sqrt{0.75 x_0^2} = \pm 0.866 \omega x_0$, evaluated with the student's own ω (3). The $\pm$ should be acknowledged: at this displacement the object may be travelling either way (1).